

COUNTEREXAMPLES TO THE WIDE-INTERVAL CASE OF CONJECTURE 4 OF GAUTSCHI AND LEOPARDI

ABSTRACT. Gautschi and Leopardi conjectured that, on $0 < \theta < \pi$, validity at $n = 1$ of a normalized comparison between consecutive Jacobi polynomials implies validity for every $n \geq 1$. Let β_* be the unique root in $(1, 2)$ of $\beta^3 + 27\beta^2 - 10\beta - 24 = 0$; numerically, $\beta_* = 1.1186397525\dots$. We show that every parameter pair $(\alpha, \beta) = (0, \beta)$ with $\beta > \beta_*$ is a counterexample: the comparison holds at $n = 1$ for every $\theta \in (0, \pi)$ but fails at $n = 2$ near $\theta = \pi$. For $\beta = 2$ we give an exact interior witness.

1. THE COUNTEREXAMPLES

For $\alpha, \beta > -1$, set

$$\tilde{P}_n^{(\alpha, \beta)}(x) = \frac{P_n^{(\alpha, \beta)}(x)}{P_n^{(\alpha, \beta)}(1)}.$$

Gautschi and Leopardi considered

$$(1) \quad \tilde{P}_n^{(\alpha, \beta)}\left(\cos \frac{\theta}{n}\right) < \tilde{P}_{n+1}^{(\alpha, \beta)}\left(\cos \frac{\theta}{n+1}\right).$$

Let $\Theta_1^{(\alpha, \beta)} \in (0, \pi)$ be defined by $P_1^{(\alpha, \beta)}(\cos \Theta_1^{(\alpha, \beta)}) = 0$. For each of the intervals $0 < \theta < \Theta_1^{(\alpha, \beta)}$ and $0 < \theta < \pi$, Conjecture 4 of Gautschi and Leopardi asserts that validity of (1) for $n = 1$ throughout the interval implies validity for every $n \geq 1$ throughout that interval [1, Conjecture 4].

Gautschi and Leopardi report numerical evidence for both branches. In the passage below, which follows the statement of Conjecture 4, their (3.6) is (1) and their (3.7) is the pair of intervals above [1, Section 3.2]:

The Matlab script of Section 2.2 is easily adapted to deal with the conjecture (3.6) for general Jacobi polynomials. When run with the same data as used to verify Conjecture 3, with $n = 100$ and $N = 1000$, similar results were obtained as in Section 3.1, i.e., a strong indication that (3.6) holds on either interval (3.7) for all $n \geq 1$ whenever $\alpha > 1$ and $\beta > -1$, while for α in the interval $(-1, 1)$ the same is true for β above a certain curve that extends from the point $(\alpha, \beta) = (-1, 0)$ down to the point $(\alpha, \beta) = (1, -1)$. For β below that curve, the conjecture fails consistently when $n = 1$.

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The sweep used to verify Conjecture 3 takes $\alpha = 0.9, 0.8, \dots, -0.9$ and, for each of these, $\beta = 20, 19, \dots, 0$ and then negative values down to -0.999 [1, Section 3.1]; thus $(\alpha, \beta) = (0, 2)$ is one of its nodes. The curve described in the passage runs from $(\alpha, \beta) = (-1, 0)$ down to $(1, -1)$ and so nowhere exceeds $\beta = 0$, whence that node lies well inside the region for which a positive indication is reported for both intervals.

At that node the failure is not marginal. In the notation of the proof of Theorem 1 with $\beta = 2$ one has $\Delta_2(\pi) = -3/16$, and Δ_2 changes sign at $\theta \approx 2.8725$, that is at $\theta \approx 0.9144\pi$. On the grid $\theta_\nu = \nu\pi/1001$, $\nu = 1, \dots, 1000$, of their routine for $0 < \theta < \pi$ with $N = 1000$, all 1000 nodes satisfy (1) at $n = 1$, the smallest margin there being 3.7×10^{-6} , while 85 of them violate it at $n = 2$, the first of these being $\nu = 916$:

$$\Delta_2(\theta_{916}) = -0.0013, \quad \Delta_2(\theta_{966}) = -0.1032, \quad \Delta_2(\theta_{1000}) = -0.1849.$$

A plausible explanation is visible in the verification core printed in their Section 2.2, where the upper limit of the θ -sweep is set by `th1=acos(-1/(2*a+1))`; while the alternative `% th1=pi;` is present but commented out. The script actually used for Conjecture 4 is described there only as an adaptation of that one and is not printed, so this explanation is circumstantial rather than documented.

Koumandos proved (1) for every $n \geq 1$ and every $\theta \in (0, \pi)$ at $(\alpha, \beta) = (1/2, 1/2), (1/2, -1/2), (-1/2, 1/2)$ [6]; that is precisely the branch refuted below, so it is true at those three pairs and false at $(\alpha, \beta) = (0, \beta)$ for every $\beta > \beta_*$ (Theorem 1). Recent work explicitly distinguishes this polynomial-value problem from the corresponding questions for Jacobi zeros [7, Section 1]. For the zeros the analogous implication is already known to fail. Extend the notation above by letting $\cos \Theta_n^{(\alpha, \beta)}$ be the largest zero of $P_n^{(\alpha, \beta)}$. Conjecture 3 of [1] asserts in the same way that validity at $n = 1$ of $n\Theta_n^{(\alpha, \beta)} < (n+1)\Theta_{n+1}^{(\alpha, \beta)}$ implies validity for every $n \geq 1$. Gautschi disproved it on a very narrow lens-shaped region inside the square $\{-1 < \alpha < -1/2, -1/2 < \beta < 0\}$ and there formulated a revised parameter domain [2]; the conjectured domains for the zeros were then extended and modified in [3, 4], with a further remark in [5]. All four of those papers concern the zeros; none revisits the polynomial-value implication (1). To our knowledge, no counterexample to that implication has previously been recorded.

Theorem 1. *Let $\beta_* \in (1, 2)$ be the unique root in that interval of*

$$q(\beta) = \beta^3 + 27\beta^2 - 10\beta - 24.$$

For every $\beta > \beta_$, the pair $(\alpha, \beta) = (0, \beta)$ is a counterexample to the $0 < \theta < \pi$ branch of Conjecture 4. More precisely,*

$$P_1^{(0, \beta)}(\cos \theta) < P_2^{(0, \beta)}\left(\cos \frac{\theta}{2}\right) \quad (0 < \theta < \pi),$$

whereas there exists $\varepsilon_\beta > 0$ such that

$$P_2^{(0,\beta)}\left(\cos \frac{\theta}{2}\right) > P_3^{(0,\beta)}\left(\cos \frac{\theta}{3}\right) \quad (\pi - \varepsilon_\beta < \theta < \pi).$$

Proof. Since $P_n^{(0,\beta)}(1) = 1$, the normalization is immaterial. From

$$P_n^{(0,\beta)}(1 - 2z) = {}_2F_1(-n, n + \beta + 1; 1; z)$$

we obtain

$$P_1^{(0,\beta)}(x) = 1 - (\beta + 2)z,$$

$$P_2^{(0,\beta)}(x) = 1 - 2(\beta + 3)z + \frac{(\beta + 3)(\beta + 4)}{2}z^2,$$

$$P_3^{(0,\beta)}(x) = 1 - 3(\beta + 4)z + \frac{3(\beta + 4)(\beta + 5)}{2}z^2 - \frac{(\beta + 4)(\beta + 5)(\beta + 6)}{6}z^3,$$

where $z = (1 - x)/2$.

Let $c = \cos(\theta/2)$ and $u = 1 - c$. Then $0 < u < 1$, and direct substitution gives

$$(2) \quad P_2^{(0,\beta)}(c) - P_1^{(0,\beta)}(\cos \theta) = u \left(\beta + 1 + \frac{\beta^2 - \beta - 4}{8}u \right).$$

The affine factor in parentheses has endpoint values

$$\beta + 1 \quad \text{and} \quad \frac{\beta^2 + 7\beta + 4}{8},$$

both positive because $\beta > \beta_* > 1$. Thus (2) is positive for every $0 < \theta < \pi$.

Now define

$$\Delta_2(\theta) = P_3^{(0,\beta)}\left(\cos \frac{\theta}{3}\right) - P_2^{(0,\beta)}\left(\cos \frac{\theta}{2}\right).$$

At the endpoint,

$$(3) \quad \Delta_2(\pi) = P_3^{(0,\beta)}\left(\frac{1}{2}\right) - P_2^{(0,\beta)}(0) = -\frac{q(\beta)}{384}.$$

Since $q(1) = -6$, $q(2) = 72$, and $q'(\beta) = 3\beta^2 + 54\beta - 10 > 0$ for $\beta \geq 1$, the number β_* is well defined and $q(\beta) > 0$ for $\beta > \beta_*$. Hence (3) is negative. Continuity therefore gives $\Delta_2(\theta) < 0$ whenever $\theta < \pi$ is sufficiently close to π . $\square$

Corollary 2. At $(\alpha, \beta) = (0, 2)$, let

$$\theta_0 = 6 \arccos \frac{7}{8}.$$

Then $0 < \theta_0 < \pi$, the comparison (1) holds at $n = 1$ for every $\theta \in (0, \pi)$, and it fails at $n = 2$ and $\theta = \theta_0$.

Proof. Put $\varphi = \arccos(7/8)$. Since $7/8 > \sqrt{3}/2$, one has $0 < \varphi < \pi/6$ and hence $0 < \theta_0 < \pi$. Moreover,

$$\cos \frac{\theta_0}{2} = \cos(3\varphi) = \frac{7}{128}, \quad \cos \frac{\theta_0}{3} = \cos(2\varphi) = \frac{17}{32}.$$

Using the formulas above,

$$P_3^{(0,2)}\left(\frac{17}{32}\right) - P_2^{(0,2)}\left(\frac{7}{128}\right) = -\frac{6785}{65536} < 0.$$

The $n = 1$ assertion follows from Theorem 1. $\square$

Remark 3 (Scope). The theorem refutes only the $0 < \theta < \pi$ branch of Conjecture 4. It does not decide the $0 < \theta < \Theta_1^{(\alpha,\beta)}$ branch, Conjecture 5 of [1], or any separate conjecture concerning Jacobi zeros. Nor is the family $\alpha = 0$ minimal: with

$$\tilde{P}_n^{(\alpha,\beta)}(1 - 2z) = {}_2F_1(-n, n + \alpha + \beta + 1; \alpha + 1; z),$$

the same endpoint computation gives

$$\tilde{P}_3^{(\alpha,\beta)}\left(\frac{1}{2}\right) - \tilde{P}_2^{(\alpha,\beta)}(0) = -\frac{49}{320}, \quad -\frac{167}{37440}, \quad -\frac{151}{6720}, \quad -\frac{5}{96}$$

at $(\alpha, \beta) = (-\frac{1}{2}, 1), (\frac{1}{4}, \frac{3}{2}), (\frac{1}{2}, 2), (1, 3)$ respectively, so the pairs of Theorem 1 form a slice of a two-dimensional region on which (1) fails at $n = 2$ near $\theta = \pi$. We work with $\alpha = 0$ because there the normalization is trivial and the witnesses are exactly rational.

REFERENCES

- [1] W. Gautschi and P. Leopardi, *Conjectured inequalities for Jacobi polynomials and their largest zeros*, Numer. Algorithms **45** (2007), no. 1–4, 217–230. doi:10.1007/s11075-007-9067-5.
- [2] W. Gautschi, *On a conjectured inequality for the largest zero of Jacobi polynomials*, Numer. Algorithms **49** (2008), 195–198. doi:10.1007/s11075-008-9207-6.
- [3] W. Gautschi, *On conjectured inequalities for zeros of Jacobi polynomials*, Numer. Algorithms **50** (2009), 93–96. doi:10.1007/s11075-008-9217-4.
- [4] W. Gautschi, *New conjectured inequalities for zeros of Jacobi polynomials*, Numer. Algorithms **50** (2009), 293–296. doi:10.1007/s11075-008-9230-7.
- [5] W. Gautschi, *Remark on “New conjectured inequalities for zeros of Jacobi polynomials”*, Numer. Algorithms **57** (2011), 511. doi:10.1007/s11075-010-9442-5.
- [6] S. Koumandos, *On a conjectured inequality of Gautschi and Leopardi for Jacobi polynomials*, Numer. Algorithms **44** (2007), no. 3, 249–253. doi:10.1007/s11075-007-9098-y.
- [7] K. Castillo and F. R. Rafaeli, *Affine scaling of Jacobi zeros: sharp orderings beyond Gautschi’s conjectures*, arXiv:2608.08258 [math.CA], 2026.